

SMART INDIA HACKATHON 2026 — TECHNICAL DOSSIER

PROBLEM STATEMENT ID: SIH26102 | MINISTRY OF STATISTICS AND PROGRAMME IMPLEMENTATION (MOSPI)

JanDrishti: Parliamentary Expenditure Intelligence & Public Works Transparency

Executive Pitch & Novelty Statement

JanDrishti is a sovereign civic intelligence, parliamentary expenditure tracking, and statutory forensic audit platform designed for the ₹11,667 Crore Members of Parliament Local Area Development Scheme (**MPLADS**).

What Makes JanDrishti Novel and Unique?

- Bicameral Line-Item Reconciliation with ₹0.00 Variance:** Existing government portals either present fragmented regional data or flat aggregated summaries. JanDrishti reconciles **778 MPs** across both the 18th Lok Sabha (542) and Rajya Sabha (236) across **36 States/UTs**, mapping **102,437 physical works** directly to **82,296 payment disbursement vouchers** and **22,377 standardized contractors** down to the exact rupee.
- Dual-Engine Zero-Downtime Topology:** Unifies cloud-native **Supabase PostgreSQL** (PostgREST HTTPS streaming) with an autonomous **Local SQLite (WAL mode)** hot-standby fallback, ensuring 100% operational continuity even in air-gapped or low-connectivity district environments.
- Deterministic Forensic AI (Zero-Hallucination Policy):** Replaces legally indefensible black-box neural networks with deterministic, peer-reviewed mathematical forensic tests: Median Absolute Deviation (MAD), Benford's Law first-digit testing, and Herfindahl-Hirschman contractor concentration indices.
- Cryptographic Role & Geographic Jurisdiction Enforcement (ABAC):** Guarantees that regional authorities (e.g., District Collectors) are strictly quarantined to their legal jurisdiction via server-validated JWT tokens (HTTP 403 Forbidden on boundary violations).

1. TECHNOLOGIES TO BE USED

Architectural Layer	Technology Stack	Exact Version / Specification	Technical Justification & Role in JanDrishti
Frontend Framework	React 19 + TypeScript	React v19.0.0 / TS ~5.7.2	Component-based, type-safe user interface; sub-millisecond reactive UI state updates across massive tabular datasets.
Build System	Vite 6	Vite v6.1.0 (Rollup engine)	High-speed Hot Module Replacement (HMR); tree-shaken code-split production bundles compiling 2,680 modules in 18s.
Styling & Interaction	Tailwind CSS + Motion	Tailwind 3.4.17 / Motion 13.1	Accessible, responsive editorial design system; GPU-accelerated micro-interactions for modals and drawers.
Geospatial Engine	D3-Geo + TopoJSON Client	d3-geo 3.1.1 / topojson 3.1.0	Vector SVG choropleth rendering all 542 Lok Sabha constituencies directly in the browser with zero external map API billing/keys .
Data Visualization	Recharts	Recharts v2.15.1	Declarative charting for expenditure velocities, sector distributions, and statistical outlier bands.
API Backend Runtime	FastAPI + Python 3.13	FastAPI 0.115 / Python 3.13	High-throughput asynchronous ASGI microservice; auto-generated OpenAPI documentation; Pydantic v2 strict type validation.
Primary Cloud Database	Supabase PostgreSQL 15	PostgreSQL 15 (AWS Tokyo)	Authoritative relational database hosting 102K works and 82K vouchers; PostgREST HTTPS streaming API; relational integrity.
Fallback Edge Database	SQLite 3 (WAL Mode)	SQLite 3.45 / 164 MB file	Local hot-standby database enabling instant offline failover, sub-5ms local queries, and rapid automated testing.
Security & Auth	HMAC-SHA256 JWT	PyJWT / Cryptographic tokens	Sovereign token-based authentication; embeds role claims and geographic jurisdiction attributes (ABAC).
Test Automation	Pytest	Pytest 9.1.1	92 unit and integration tests verifying API contracts, authentication, failover, and boundary enforcement (100% pass rate).
Hardware & Deployment	Containerized Microservices	Docker / Alpine Linux / Nginx	Deployable on NIC MeghRaj cloud, AWS ECS, or local edge hardware; minimal footprint (runs comfortably on 2 vCPUs and 2GB RAM).

2. METHODOLOGY AND PROCESS FOR IMPLEMENTATION

A. End-to-End System Process Flowchart



B. Implementation Methodology & Forensic Algorithms

- 1. **Canonical Entity Resolution:** Public government contractor records often contain typos and inconsistent naming (`M/s ABC Construction` , `ABC Construction Ltd` , `A.B.C. Const`). Our normalization pipeline executes string stripping, legal honorific pruning, and Levenshtein distance clustering to map 34,000 raw strings into 22,377 canonical vendor entities.
- 2. **Median Absolute Deviation (MAD) Robust Outlier Scoring:** Traditional mean/standard deviation models are easily distorted by extreme government tenders. JanDrishti implements the Rousseeuw & Croux scale estimator: $Z_{\text{Modified}} = \frac{|X_i - \text{Median}|}{\text{MAD}}$ Works exceeding $Z > 3.0$ within their specific sector (e.g., Rural Roads vs. Hospital Equipment) are flagged for human inspection.
- 3. **Benford's Law First-Digit Distribution Test:** Natural financial ledgers follow a logarithmic distribution of first digits: $P(d) = \log_{10}\left(1 + \frac{1}{d}\right)$ In procurement, artificial spikes in leading digits (such as digit `4` for ₹4,95,000) indicate split-billing designed to bypass the ₹5 Lakh statutory e-tendering threshold.
- 4. **Herfindahl-Hirschman Index (HHI) Vendor Reliance:** Calculates contractor market share across a single constituency: $HHI = \sum_{i=1}^n s_i^2$ If a single contractor captures $>60\%$ of constituency disbursements, an analytical risk flag is issued.

3. FEASIBILITY AND VIABILITY ANALYSIS

A. Technical, Operational, and Economic Feasibility

- **Technical Feasibility (100% Proven):** The system is fully operational. We have built, tested, and validated the working prototype against 102,437 physical works and 82,296 vouchers. The backend test suite executes 92 automated Pytests in 1.48 seconds; the frontend builds cleanly with zero TypeScript errors.
- **Operational Feasibility:** Aligns directly with existing administrative hierarchies under the **MoSPI MPLADS Guidelines (2023)**. District Collectors, State Nodal Officers, and Ministry Administrators receive role-tailored dashboards matching their statutory duties (approving DPRs, tracking utilization tranches, sanctioning milestones).
- **Economic Viability:** Built entirely on open-source, sovereign technologies (React, FastAPI, PostgreSQL, SQLite, D3-Geo). Zero proprietary runtime software licenses or Google Maps API charges. By surfacing **₹7,720 Crore** in stagnant unspent balances and preventing duplicate DPR funding, the platform provides immense financial return on investment.

B. Potential Challenges, Risks & Strategic Mitigation

Potential Challenge / Risk	Risk Severity	Real-World Impact	JanDrishti Strategic Mitigation
Data Opaqueness & Non-Uniformity	HIGH	Government portals export mismatched formats between Lok Sabha (granular line-items) and Rajya Sabha (state aggregates).	Transparent Granularity Disclosure: We explicitly document bicameral differences in the UI rather than synthesizing fake data. Reconciled with ₹0.00 mathematical precision.
Network & Cloud Outages in Field	MEDIUM	District Collectors in remote areas face intermittent internet, failing cloud API calls.	Dual-Engine Architecture: Automatic failover from Supabase PostgreSQL to local SQLite (WAL mode) database on local edge servers with zero interruption.
Cross-Jurisdiction Data Leakage	HIGH	A state or district official accessing another region's confidential project estimates.	Cryptographic ABAC Enforcement: User tokens sign geographic claims. Backend endpoints strictly enforce <code>WHERE district_id = :claim</code> . URL tampering triggers HTTP 403.
Defamation & Political Resistance	HIGH	Statistical flags being misconstrued as legal accusations of corruption against MPs.	Neutral Governance Terminology: All signals are strictly labeled "Analytical Risk Indicator — Requires Review" under CAG audit standards, recognizing legitimate terrain factors.
Data Freshness / Sync Lag	MEDIUM	MoSPI does not currently provide live WebSockets for instant transaction push.	Modular Ingestion Engine: Automated batch staging interface ready to transition immediately to PFMS webhooks once official government API sandboxes open.

4. IMPACT AND BENEFITS

A. Impact on Target Audiences

- For Citizens:** Replaces bureaucratic opacity with 1-click discovery. Citizens can search their home constituency, verify village project completion, and submit geo-tagged photo feedback.
- For Members of Parliament (MPs):** Real-time visibility into portfolio execution velocity, pending sanctions with the District Collector, and unspent balances to prevent fund stagnation.
- For District Collectors & Authorities:** Dedicated workflow workspace to inspect DPRs, verify compliance with MoSPI guidelines, approve construction milestones, and release vendor tranches.
- For the Comptroller and Auditor General (CAG) & Auditors:** Instant forensic triage. 1,831 pre-computed statistical anomaly flags and 1-click printable PDF audit dossiers.
- For the Central Ministry (MoSPI):** Pan-India bicameral oversight across all 36 States with automated inter-state utilization leaderboards and early warnings on slow tranches.

B. Multi-Dimensional Benefits of the Solution

- Social & Democratic Accountability:** Restores citizen trust in public infrastructure spending. Transforms passive taxpayers into active participants in local governance.
- Economic Efficiency & Leakage Prevention:** Helps unlock ₹7,720.09 Crore of stagnant unspent public capital. Prevents duplicate billing and contractor cartelization across neighboring districts.
- Administrative Velocity:** Replaces cumbersome physical paper files and scattered spreadsheets with instantaneous digital milestone sanctioning and exportable case dossiers.
- Environmental & Sustainable Development:** Tracks green capital assets (solar street lighting, water conservation ponds, bio-toilets) mapped across regional developmental zones.

5. RESEARCH AND REFERENCES

The design and mathematical models of JanDrishti are grounded in statutory governance frameworks and peer-reviewed forensic literature:

- Ministry of Statistics and Programme Implementation (MoSPI):**
Guidelines on Members of Parliament Local Area Development Scheme (MPLADS), Government of India, Revised Edition (2023).
Reference: Statutory fund release guidelines, eligible sector works, administrative sanction thresholds, and nodal district obligations.
- Comptroller and Auditor General of India (CAG):**
Performance Audit Report on Members of Parliament Local Area Development Scheme (MPLADS), Report No. 31 of 2010-11 and Report No. 19 of 2018.
Reference: Documented findings on unspent fund accumulation, contractor concentration, execution delays, and monitoring gaps.
- Forensic Financial Analytics (Benford's Law):**
Nigrini, Mark J. (2012). *Benford's Law: Applications for Forensic Accounting, Auditing, and Fraud Detection*. John Wiley & Sons.

Reference: Mathematical foundation for assessing first-digit probability anomalies in multi-order-of-magnitude procurement disbursement vouchers.

4. **Robust Statistics & Scale Estimators:**

Rousseeuw, Peter J., and Croux, Christophe (1993). *Alternatives to the Median Absolute Deviation*. Journal of the American Statistical Association, 88(424), 1273-1283.

Reference: Utilization of the normal distribution scale factor (0.6745) for outlier-resistant Z-score derivation in public expenditure datasets.

5. **Market Concentration & Competition Forensics:**

U.S. Department of Justice & Federal Trade Commission (2010). *Horizontal Merger Guidelines (Herfindahl-Hirschman Index - HHI)*.

Reference: Formulation for measuring vendor monopolization ($\$HHI > 2500\$$) within localized district procurement markets.

6. **Sovereign Open Data Sources:**

- Sansad.in: *Official Members of Parliament Directory (18th Lok Sabha & Rajya Sabha)*.
- Election Commission of India (ECI): *Parliamentary Constituency Boundaries and Delimitation Gazette*.
- Survey of India: *National Geo-Spatial Vector Repository*.

Verification and Evaluation Summary

- **Working Prototype:** Live on `http://localhost:3000` with hidden evaluation dossier on `/docs`.
- **Backend API Daemon:** Active on `http://127.0.0.1:8000` with 92 passing automated tests.
- **Sovereign Codebase:** 100% auditable, zero proprietary cloud lock-in, ready for immediate government pilot deployment.